Module I: Basics of R Programming

Reading and Writing Data in R - Part II

Readings: Chapter 8. and 9. in *R in Programming for Data Science*

Textual Formats

- `dumping` and dputing are useful because the resulting textual format is edit-able, and in the case of corruption, potentially recoverable.
- `Unlike` writing out a table or csv file, `dump` and `dput` preserve the *metadata* (sacrificing some readability), so that another user doesn't have to specify it all over again.
- `Textual` formats can work much better with version control programs like subversion or git which can only track changes meaningfully in text files
- Textual formats can be longer-lived; if there is corruption somewhere in the file, it can be easier to fix the problem
- Textual formats adhere to the "Unix philosophy"
- Downside: The format is not very space-efficient

dput-ting R Objects

Another way to pass data around is by deparsing the R object with dput and reading it back in using `dget`.

```
> y <- data.frame(a = 1, b = "a")
> dput(y)
structure(list(a = 1,
               b = structure(1L, .Label = "a",
                             class = "factor")),
          .Names = c("a", "b"), row.names = c(NA, -1L),
          class = "data.frame")
> dput(y, file = "y.R")
> new.y <- dget("y.R")
> new.y
   a  b
1  1  a
```

Dumping R Objects

Multiple objects can be deparsed using the dump function and read back in using `source`.

```
> x <- "foo"
> y <- data.frame(a = 1, b = "a")
> dump(c("x", "y"), file = "data.R")
> rm(x, y)
> source("data.R")
> y
  a  b
1 1  a
> x
[1] "foo"
```

Interfaces to the Outside World

Data are read in using *connection* interfaces. Connections can be made to files (most common) or to other more exotic things.

- `file`, opens a connection to a file
- `gzfile`, opens a connection to a file compressed with gzip
- `bzfile`, opens a connection to a file compressed with bzip2
- `url`, opens a connection to a webpage

File Connections

```
> str(file)
function (description = "", open = "", blocking = TRUE,
          encoding = getOption("encoding"))
```

- `description` is the name of the file
- `open` is a code indicating
  - "r" read only
  - "w" writing (and initializing a new file)
  - "a" appending
  - "rb", "wb", "ab" reading, writing, or appending in binary mode (Windows)

Connections

In general, connections are powerful tools that let you navigate files or other external objects. In practice, we often don't need to deal with the connection interface directly.

```
con <- file("foo.txt", "r")
data <- read.csv(con)
close(con)
```

is the same as

```
data <- read.csv("foo.txt")
```

Reading Lines of a Text File

```
> con <- gzfile("words.gz")
> x <- readLines(con, 10)
> x
 [1] "1080"      "10-point" "10th"      "11-point"
 [5] "12-point" "16-point" "18-point" "1st"
 [9] "2"         "20-point"
```

`writeLines` takes a character vector and writes each element one line at a time to a text file.

Reading Lines of a Text File

`readLines` can be useful for reading in lines of webpages

```
## This might take time
con <- url("http://www.jhsph.edu", "r")
x <- readLines(con)
> head(x)
[1] "<!DOCTYPE HTML PUBLIC \"-//W3C//DTD HTML 4.0 Transitional//EN\">"
[2] ""
[3] "<html>"
[4] "<head>"
[5] "\t<meta http-equiv=\"Content-Type\" content=\"text/html;charset=utf-8
```